

ALFIDO TECH INTERNSHIP TASK REPORT

Project Title: Zomato Data Analysis and Visualizations Project

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Internship Domain: Data Science

Task Name: Exploratory Data Analysis (EDA)

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Project Objective

The core objective of this project is to provide an end-to-end data science experience: from data cleaning and exploratory analysis to model building, evaluations, and deployment. Specifically, this task analyzes the Zomato dataset using Python to clean messy restaurant data and uncover meaningful business insights regarding restaurant ratings, operational costs, and customer habits.

Data Cleaning & Preparation Process

The dataset was successfully loaded, and libraries like pandas, numpy, matplotlib, and seaborn were utilized. Text/string anomalies in the rating and cost columns were removed to transform them into numeric formats for downstream statistics. Duplicate values were systematically dropped to maintain data integrity.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from wordcloud import WordCloud
import warnings
warnings.filterwarnings('ignore')

print("Libraries loaded successfully!")

# Load and drop duplicates
df = pd.read_csv('zomato.csv')
df.isnull().sum()
df.drop_duplicates(inplace=True)

# Clean rating text like '
Top floor' or 'NEW' into missing/NaN values
df['rate'] = df['rate'].apply(clean_rating)
print("Data types after cleaning:")
print(df[['approx_cost(for two people)', 'rate']].dtypes)
```

Data types after cleaning:

approx_cost(for two people) : float64

rate : float64

First 5 Cleaned Rows Sample

Name	Approx Cost (for two people)	Rate (out of 5)
Jalsa	800.0	4.1
Spice Elephant	800.0	4.1
San Churro Cafe	800.0	3.8
Addhuri Udupi Bhojana	300.0	3.7
Grand Village	600.0	3.8

Data Analysis & Visualizations Insights

Exploratory Data Analysis was executed to study trends between price points, customer feedback ratings, and online delivery behaviors.

Distribution of Restaurant Ratings & Cost Analysis

[Histogram: Distribution of Restaurant Ratings]

Shows a normal bell-curve distribution centered around a 3.7 to 4.0 average score rating.

[Scatter Plot: Cost for Two vs. Rating]

Visualizes how budget levels correlate to rating frequencies across common price ranges up to ₹3000.

Online Ordering Impact Analysis

Analysis was conducted on how online ordering features affect general restaurant counts and their corresponding average ratings.

Key Insights from Online Order Status:

- **Volume Analysis:** Outlines customer frequencies for 'Yes' vs. 'No' delivery support.
- **Rating Variance:** Quantifies shifts in customer satisfaction scores depending on the available online fulfillment.

Customer Feedback & Review Mining Examples

Text logs show rich customer review data ranging across sentiment thresholds:

- "If you are in the mood for some super North Indian food, you have to hit this place up..." (Rated 4.0 / 5.0)
- "One fine HALLOWEEN'18 night @ this most amazing place... food here is amazing and service is super quick..."
- "Worst customer service... when I called the restaurant they did not seem to care about it."
- "...will never ever order from them again. We ordered the ajwain fish..."